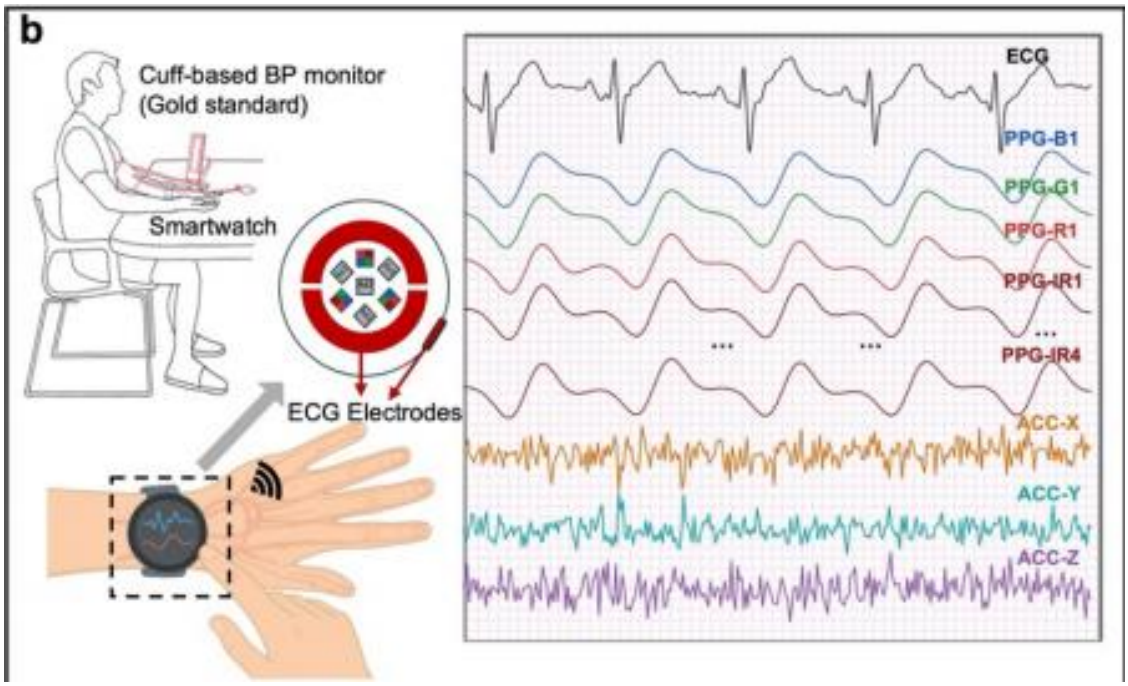
Protocol and Data Details

With the approval of the Institutional Review Board of Songshan Lake Central Hospital with No. DSY-IRB-2025-06, we recruited 5252 participants for a single blood pressure experiment (S-BP) and 451 participants for follow-up pressure experiment (F-BP) lasting approximately four months.



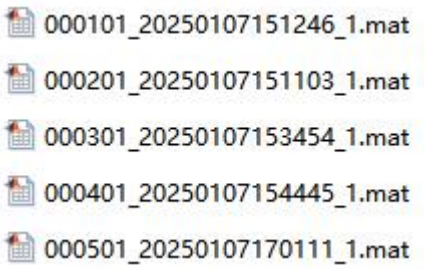
The smartwatch utilized in the study was a prototype provided by Honor Device Co., Ltd., that simultaneously records three-axis acceleration, ECG, and MWPPG signals. The device integrates an active standard lead-I ECG module and a 16-channel MSPPG sensor array, comprising four spatially distributed photodetector (PD) units arranged along the radial (Rd) and axial (Ax) directions. The PD is surrounded by LEDs of four wavelengths, such as 940 nm (infrared, PPG_IR), 660 nm (red, PPG_R), 525 nm (green, PPG_G) and 455 nm (blue, PPG_B), to provide comprehensive and convergent spectral coverage. The system supports both multi-source single-detector (4-channel) and multi-source multi-detector (16-channel) configurations. The sampling frequency was set at 100 Hz for three-axis acceleration signals and 500 Hz for ECG and MWPPG signals. (**Figure above**)

The experimental procedure consisted of the following standardized steps: **(i)** Preparation: Participants were asked to remain seated and relaxed for 5 minutes prior to measurement. **(ii)** Baseline BP measurement: According to standard clinical protocols], two consecutive BP readings were obtained from the left upper arm using an automated electronic sphygmomanometer. If the difference between the two readings was ≤ 4 mmHg, their mean was recorded as the baseline BP (SBP_baseline and DBP_baseline); otherwise, a third measurement was performed.

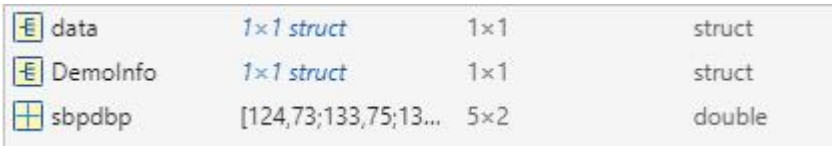
(iii) Signal acquisition: Participants wore the smartwatch on the left wrist while continuous ECG and MWPPG signals were recorded for 3 minutes, including 1 minute of 4-channel MWPPG followed by 2 minutes of full 16-channel MWPPG. **(iv)** Snapshot reference BP measurement: Immediately after signal acquisition, a reference BP reading (SBP_ref and DBP_ref) was obtained from the same arm using the same standardized protocol. Steps **(iii)** and **(iv)** were repeated across four consecutive rounds.

For S-BP, each participant underwent at least four recordings on one day. For F-BP, each participant underwent at least four recordings on day D, D+7, D+14, D+21, D+28, D+56, D+84 and D+112 for 8 different days.

Data File Descriptions



SubjectID: The first six digits (000101,000201...)
Data-collecting-time:The numbers in the middle part(20250107151246..)
Number of collections: The last number (for S-BP, this number will always be 1, for F-BP, this number will be D (1), D+7 (2)... D+112 (8)



data	1×1 struct	1×1	struct
DemoInfo	1×1 struct	1×1	struct
sbpdbp	[124,73;133,75;13...	5×2	double

For each subject, there are three “mat” files. 1.data 2.Demoinfo3.sbpdbp

Data: it contains raw waveform data.

Four consecutive rounds: round1; round2; round3; round4
Each Round: channel4_1: first 1 minute of 4-channel MWPPG
channel16_1：second 1 minute of 16-channel MWPPG
channel16_2：last 1 minute of 16-channel MWPPG

round1	1×1 struct	1×1	struct
channel16_1	1×1 struct	1×1	struct
channel16_2	1×1 struct	1×1	struct
channel4_1	1×1 struct	1×1	struct
Acc_x_100	1×6550 dou...	1×6550	double
Acc_y_100	1×6550 dou...	1×6550	double
Acc_z_100	1×6550 dou...	1×6550	double
PPG3_100	1×32310 do...	1×32310	double
PPG4_100	1×32310 do...	1×32310	double
PPG5_100	1×32310 do...	1×32310	double
PPG6_100	1×32310 do...	1×32310	double
ECG_500	1×32340 int...	1×32340	int32
round2	1×1 struct	1×1	struct
channel16_1	1×1 struct	1×1	struct
channel16_2	1×1 struct	1×1	struct
channel4_1	1×1 struct	1×1	struct
round3	1×1 struct	1×1	struct
channel16_1	1×1 struct	1×1	struct
channel16_2	1×1 struct	1×1	struct

channel4_1 data

Mat key	Meaning	Sampling Frequency
Acc_x_100	X-axis accelerometer	100Hz
Acc_y_100	Y-axis accelerometer	100Hz
Acc_z_100	Z-axis accelerometer	100Hz
PPG3_100	IR PPG	500Hz
PPG4_100	R PPG	500Hz
PPG5_100	G PPG	500Hz
PPG6_100	B PPG	500Hz
ECG_500	ECG	500Hz

channel16_1 data

Mat key	Meaning	Sampling Frequency
Acc_x_100	X-axis accelerometer	100Hz
Acc_y_100	Y-axis accelerometer	100Hz
Acc_z_100	Z-axis accelerometer	100Hz
PPG1_100	First IR PPG	500Hz
PPG2_100	Second IR PPG	500Hz
PPG3_100	Third IR PPG	500Hz
PPG4_100	Fourth IR PPG	500Hz
PPG5_100	First R PPG	500Hz
PPG6_100	Second R PPG	500Hz
PPG7_100	Third R PPG	500Hz
PPG8_100	Fourth R PPG	500Hz
PPG9_100	First G PPG	500Hz
PPG10_100	Second G PPG	500Hz
PPG11_100	Third G PPG	500Hz

PPG12_100	Fourth G PPG	500Hz
PPG13_100	First B PPG	500Hz
PPG14_100	Second B PPG	500Hz
PPG15_100	Third B PPG	500Hz
PPG16_100	Fourth B PPG	500Hz
ECG_500	ECG	500Hz

DemoInfo: it contains the information of the subject.

age	'24'	1×2	char
height	'180'	1×3	char
medicationHis	'否'	1×1	char
medicationToday	'否'	1×1	char
medicationType	''	0×0	char
sex	'男'	1×1	char
testCount	'1'	1×1	char
userName	'000101'	1×6	char
weight	'90'	1×2	char

Age, in years
Height: in cm
medicationHis: If the participant has a history of hypertension? Yes(是) or No(否)
medicationToday: if the participant takes any medication? Yes(是) or No(否)
medicationType: the name of the drug
Sex: male(男) or female (女)
TestCount: number of collections
UserName: User ID
Weight: in kg

sbpdbp: BP values of all rounds

	1	2
1	124	73
2	133	75
3	131	80
4	138	79
5	132	79

1: Baseline BP
2-5: BP values of four rounds